

BREWTON 3 NNE, AL, USA

WMO: 722277

Lat: **31.1449N**Lon: **87.0518W**Elev: **52**StdP: **100.70**Time Zone: **-6.00 (NAC)**Period: **08-19**WBAN: **63899****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 6.1	(k) 9.0	(l) 5.6	(m) 9.2	(n) 1.7	(o) 10	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.1	(c) 34.1	(d) N/A	(e) 33.1	(f) N/A	(g) 32.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 1.6	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 4.7	(b) 4.0	(c) 3.5	(e) -7.0	(f) 35.2	(g) 2.2	(h) 1.3	(i) -8.5	(j) 36.2	(k) -9.8	(l) 37.0	(m) -11.0	(n) 37.8	(o) -12.6	(p) 38.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	9.1	11.9	15.0	18.6	22.5	26.0	26.6	26.4	24.7	19.3	13.4	11.4	
	DBStd	7.32	5.77	5.33	4.72	3.32	2.89	1.69	1.41	1.53	2.35	4.18	4.66	5.40	
	HDD10.0	207	87	38	13	1	0	0	0	0	0	1	18	49	
	HDD18.3	1061	288	186	123	36	7	0	0	0	1	40	159	221	
	CDD10.0	3419	61	93	167	260	388	481	515	509	441	288	121	94	
	CDD18.3	1229	3	8	19	46	136	231	257	251	191	69	12	7	
	CDH23.3	10413	6	28	133	416	1209	2114	2275	2087	1498	566	62	18	
	CDH26.7	4061	0	0	13	66	423	946	991	880	590	147	6	0	
Wind	WSAvg	1.4	1.7	1.8	1.8	1.6	1.2	1.1	1.0	1.0	1.0	1.2	1.2	1.6	
Precipitation	PrecAvg	1619	131	129	164	118	125	160	184	154	137	86	124	114	
	PrecMax	2422	319	259	402	283	406	393	485	373	640	380	330	318	
	PrecMin	1099	31	31	20	13	5	27	42	42	23	0	30	24	
	PrecStd	267	57	59	95	73	82	82	78	79	116	78	68	52	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	25.8	28.2	29.8	33.3	35.6	34.8	34.9	34.0	31.7	27.3	25.1	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	22.4	24.2	26.5	28.5	31.7	34.1	33.6	33.5	32.6	29.9	25.2	23.7	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	20.6	22.3	24.9	27.3	30.4	32.9	32.7	32.5	31.5	28.4	23.5	22.2	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	18.7	20.7	23.0	26.0	29.3	31.7	31.6	31.3	30.3	26.7	21.8	20.4	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.4	13.1	13.0	12.2	10.8	10.1	9.8	10.7	12.9	13.6	11.1	
		MCDBR	12.1	12.1	13.2	14.1	13.5	12.6	11.5	11.4	12.0	13.6	13.4	10.8	
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Clear-Sky Solar Irradiance	taub	0.326	0.343	0.365	0.393	0.436	0.466	0.525	0.508	0.442	0.378	0.349	0.332	
		taud	2.540	2.494	2.426	2.351	2.264	2.236	2.085	2.129	2.289	2.445	2.470	2.532	
Ebn at Noon		905	919	918	899	858	828	778	787	830	868	867	878		
Edh at Noon		85	98	112	125	137	141	163	154	127	101	91	82		
All-Sky Solar Radiation	RadAvg	2.96	3.53	4.78	5.82	6.41	6.17	5.90	5.51	5.07	4.41	3.36	2.58		
	RadStd	0.21	0.46	0.40	0.48	0.38	0.50	0.38	0.37	0.34	0.49	0.30	0.26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+97

Nomenclature: See separate page